

Input-to-State Stability for Switched Systems with Unstable Subsystems: A Hybrid Lyapunov Construction

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Outline

- ❖ Problem formulation and preliminaries
- ❖ Assumptions and sufficient condition
- ❖ Proof: auxiliary timer and hybrid ISS Lyapunov function
- ❖ Simulation
- ❖ Conclusion

Problem Formulation

Switched system

$$\dot{x} = f_{\sigma}(x, u)$$

- $\dot{x} = f_p(x, u)$, $p \in \mathcal{P}$ is a family of dynamical systems
- $\mathcal{P}$ is the **index set** (arbitrary)
- $\sigma: [0, \infty) \rightarrow \mathcal{P}$ is the **switching signal** (piecewise constant, right-continuous)
- x doesn't change at switches

Input-to-State Stability

A dynamical system is **input-to-state stable (ISS)** if

$$|x(t)| \leq \beta(|x(0)|, t) + \gamma(\|u\|_t) \quad \forall t$$

for all initial condition $x(0)$ and all input u .

- $\beta \in \mathcal{KL}$: increasing in x , $\beta(0, t) = 0$, decreasing in t and $\lim_{t \rightarrow \infty} \beta(x, t) = 0$
- $\gamma \in \mathcal{K}_\infty$: increasing, unbounded and $\gamma(0) = 0$
- $\|\cdot\|_t$ denotes the essential supremum norm on $[0, t]$
- For an autonomous system, ISS is equivalent to **global asymptotic stability (GAS)**

Problem Formulation

Switched system with **non-ISS** subsystems

$$\dot{x} = f_{\sigma}(x, u)$$

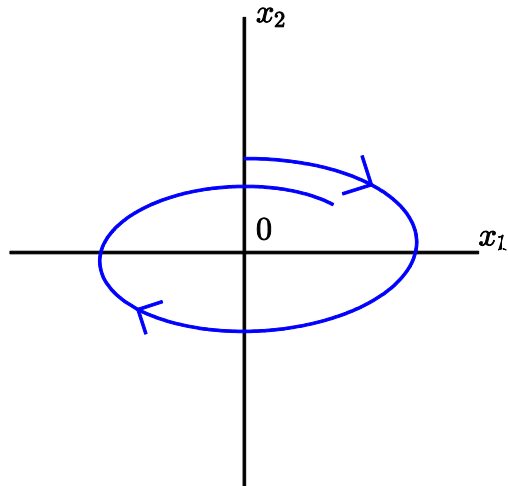
- The index set $\mathcal{P} = \mathcal{P}_s \cup \mathcal{P}_u$
 - $\mathcal{P}_s$ contains the indices of **ISS** subsystems
 - $\mathcal{P}_u$ contains the indices of **non-ISS** subsystems

Question: When is the switched system ISS?

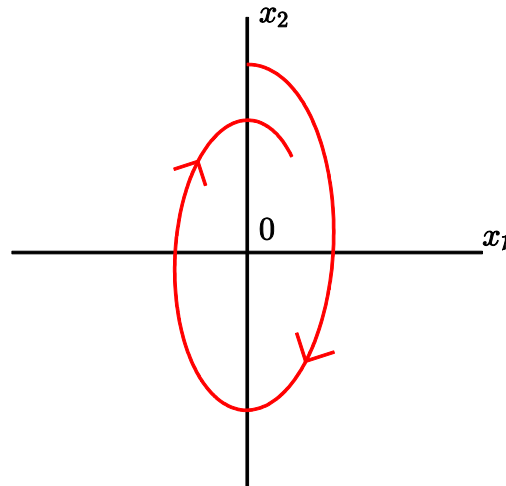
Obstacles: 1) Non-ISS subsystems
2) Switching

Stability Issue of Switching

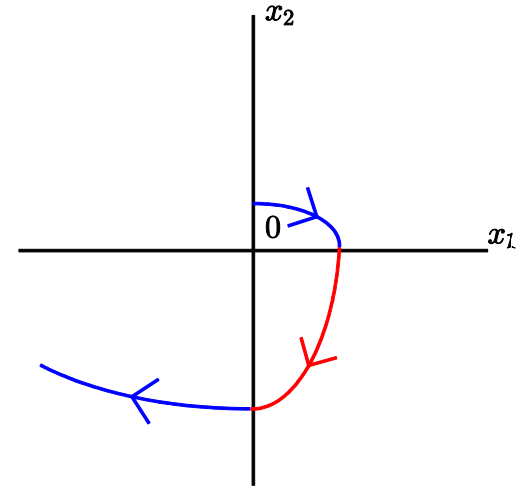
Example:



$$\dot{x} = f_1(x)$$



$$\dot{x} = f_2(x)$$



$$\dot{x} = f_\sigma(x)$$

Asymptotic stability of subsystems are **not sufficient**
for stability of the switched system

Slow Switching

Average dwell-time condition [Hespanha-Morse-99]

$$N(t_2, t_1) \leq N_0 + \frac{t_2 - t_1}{\tau_a} \quad \forall t_2 \geq t_1 \geq 0$$

- $N(t_2, t_1)$ is the number of switches on $(t_1, t_2]$
- τ_a is called the **average dwell-time**
- N_0 is the chatter bound
 - $N_0 = 0$ - no switch: cannot switch for any $t_2 - t_1 < \tau_a$
 - $N_0 = 1$ - dwell-time: can only switch once for any $t_2 - t_1 < \tau_a$

Slow Switching

Average dwell-time condition [Hespanha-Morse-99]

$$N(t_2, t_1) \leq N_0 + \frac{t_2 - t_1}{\tau_a} \quad \forall t_2 \geq t_1 \geq 0$$

Theorem

$$\begin{array}{llll} \alpha_1(|x|) \leq V_p(x) \leq \alpha_2(|x|) & \forall p & & \\ \frac{\partial V_p(x)}{\partial x} f_p(x) \leq -\lambda V_p(x) & \forall p & \Rightarrow & \dot{x} = f_\sigma(x) \text{ is GAS if} \\ V_p(x) \leq \mu V_q(x) & \forall p, q & & \tau_a > \frac{\log \mu}{\lambda} \end{array}$$

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Assumptions on dynamics

Switched system with non-ISS subsystem

$$\dot{x} = f_{\sigma}(x, u)$$

■ Uniform ISS Lyapunov-type Constraint

1. $\exists \alpha_1, \alpha_2 \in \mathcal{K}_{\infty}$ s.t. $\alpha_1(|x|) \leq V_p(x) \leq \alpha_2(|x|) \quad \forall p \in \mathcal{P}$

2. $\exists \phi \in \mathcal{K}_{\infty}, \lambda_s, \lambda_u > 0$ s.t.

$$|x| \geq \phi(|u|) \Rightarrow$$

$$\begin{cases} \frac{\partial V_{p_s}(x)}{\partial x} \cdot f_{p_s}(x, u) \leq -\lambda_s V_{p_s}(x) & \forall p_s \in \mathcal{P}_s \\ \frac{\partial V_{p_u}(x)}{\partial x} \cdot f_{p_u}(x, u) \leq \lambda_u V_{p_u}(x) & \forall p_u \in \mathcal{P}_u \end{cases}$$

due to ISS
[Praly-Wang-96]

3. $\exists \mu > 1$ s.t. $V_p(x) \leq \mu V_q(x) \quad \forall p, q \in \mathcal{P}$

forward complete
[Angeli-Sontag-99]

discussion in
[Vu-Chatterjee-L-07]

Assumptions on switching

Switched system with non-ISS subsystem

$$\dot{x} = f_{\sigma}(x, u)$$

- Average dwell-time constraint [Hespanha-Morse-99]

$$\exists \tau_a > 0, N_0 \geq 0 \text{ s.t.}$$

$$N(t_2, t_1) \leq N_0 + \frac{t_2 - t_1}{\tau_a} \quad \forall t_2 \geq t_1 \geq 0$$

- Time-ratio constraint [Zhai et al.-00]

$$\exists \rho \in [0, 1), T_0 \geq 0 \text{ s.t.}$$

$$T_u(t_2, t_1) \leq T_0 + \rho(t_2 - t_1) \quad \forall t_2 \geq t_1 \geq 0$$

- $T_u(t_2, t_1)$ is the total activation time of non-ISS subsystems on $(t_1, t_2]$, i.e., when $p \in \mathcal{P}_u$

Sufficient condition for ISS

Switched system with non-ISS subsystem

$$\dot{x} = f_{\sigma}(x, u)$$

Theorem [Müller-L-2012, Theorem 2]

Suppose all previous assumptions hold. Then the switched system is ISS if

$$\lambda_s > \frac{\ln(\mu)}{\tau_a} + \rho(\lambda_s + \lambda_u) =: \gamma$$

Interpretation: average rate of exponential decay

$$(1 - \rho)\lambda_s - \rho\lambda_u - \frac{\ln(\mu)}{\tau_a} > 0,$$

due to ISS

due to non-ISS

due to switching

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Sketch of Proof

- ISS Lyapunov-type functions of subsystems $V_p(x)$, $p \in \mathcal{P}$
- Concatenation of ISS Lyapunov-type functions $V_\sigma(x)$

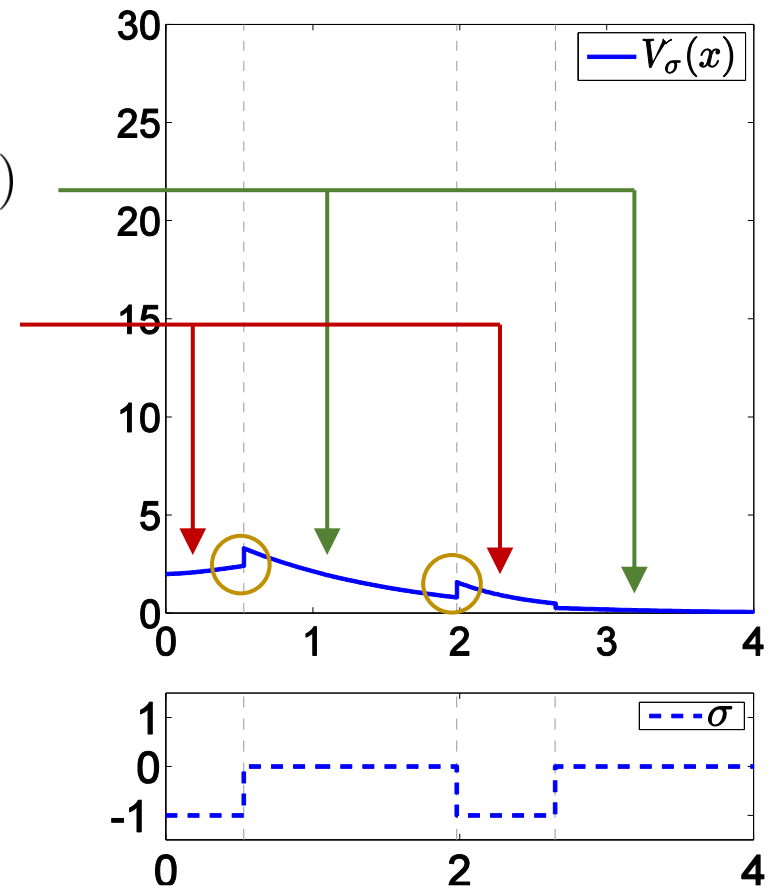
...

$$\begin{cases} \frac{\partial V_{p_s}(x)}{\partial x} \cdot f_{p_s}(x, u) \leq -\lambda_s V_{p_s}(x) \\ \frac{\partial V_{p_u}(x)}{\partial x} \cdot f_{p_u}(x, u) \leq \lambda_u V_{p_u}(x) \end{cases}$$

...

$$V_p(x) \leq \mu V_q(x)$$

...



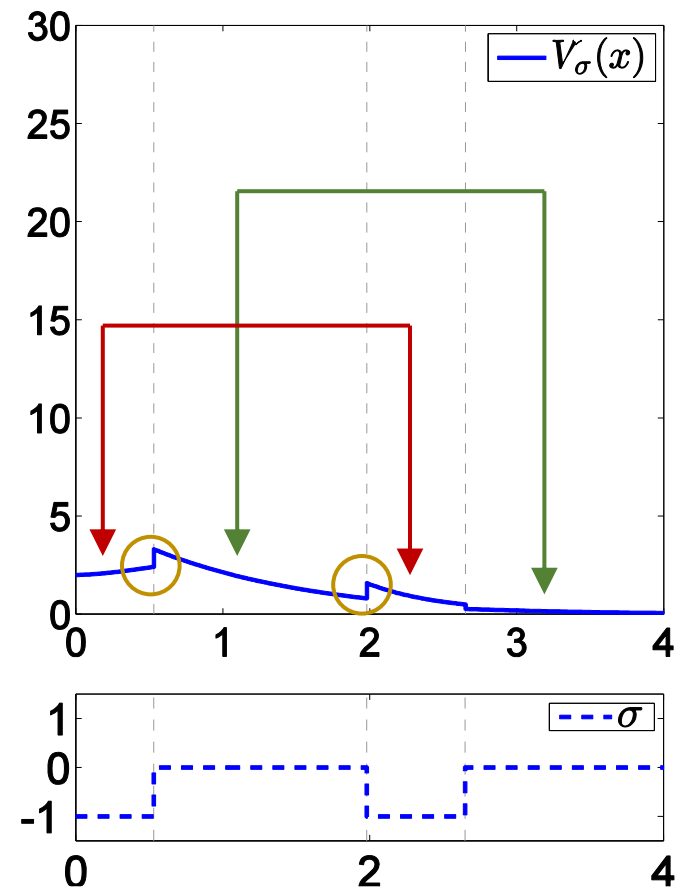
Sketch of Proof

- ISS Lyapunov-type functions of subsystems $V_p(x)$, $p \in \mathcal{P}$
- Concatenation of ISS Lyapunov-type functions $V_\sigma(x)$

$$V_{Goal}(x) = V_\sigma(x) \cdot W(x, \sigma)$$

$W(x, \sigma)$ should be

- decrease when non-ISS
- decrease at switches
- Increase when ISS



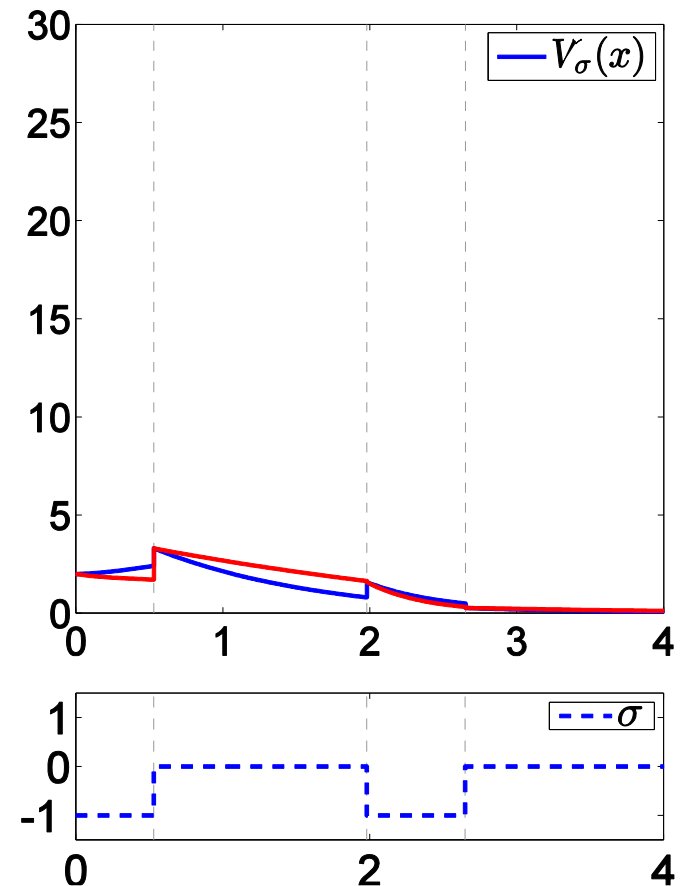
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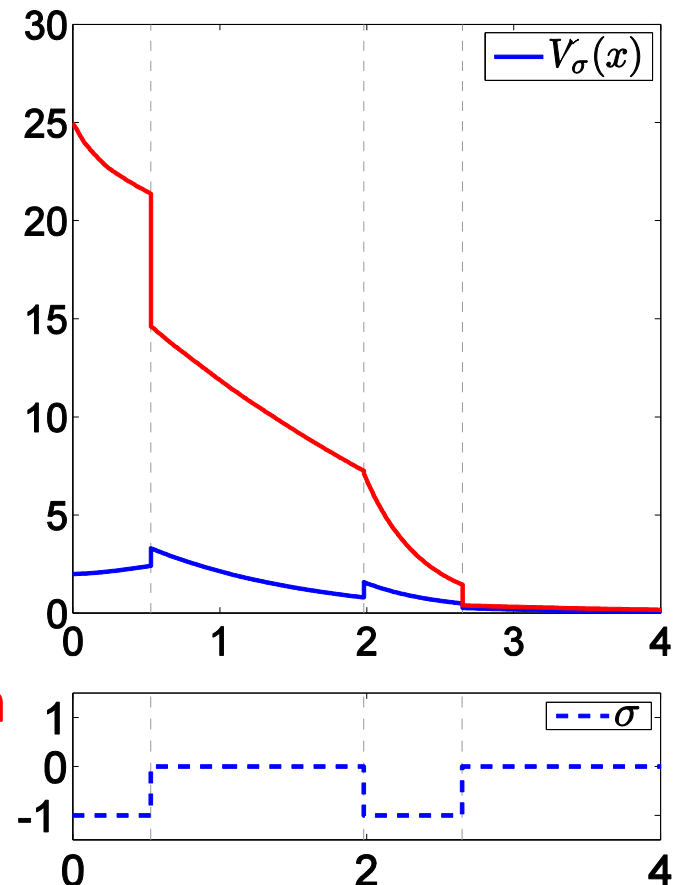
$$V_{Goal}(x) = V_\sigma(x) \cdot W(x, \sigma)$$

$W(x, \sigma)$ should be

- decrease when non-ISS
- decrease at switches
- Increase when ISS

How?

Auxiliary timer τ and hybrid system



Auxiliary Timer

An auxiliary **timer** $\tau \in [0, \Gamma]$ that models the constraints on the switching

$$\begin{cases} \dot{\tau} \in [0, \gamma], & \text{if } \tilde{\sigma} \in \mathcal{P}_s \\ \dot{\tau} = \gamma - (\lambda_s + \lambda_u), & \text{if } \tilde{\sigma} \in \mathcal{P}_u \end{cases}$$
$$\tau^+ = \tau - \ln(\mu)$$

- τ increases when an ISS subsystem is active
- τ decreases when a non-ISS subsystem is active
- τ decreases when a switch occurs

Hybrid System

An auxiliary **timer** $\tau \in [0, \Gamma]$ that models the constraints on the switching

$$\begin{cases} \dot{\tau} \in [0, \gamma], & \text{if } \tilde{\sigma} \in \mathcal{P}_s \\ \dot{\tau} = \gamma - (\lambda_s + \lambda_u), & \text{if } \tilde{\sigma} \in \mathcal{P}_u \end{cases}$$
$$\tau^+ = \tau - \ln(\mu)$$

A **hybrid system** with the state $z = (\tilde{x}, \tilde{\sigma}, \tau)$

Proposition 1

Suppose all assumptions hold. For any solution to the switched system, there is a complete solution to the hybrid system s.t.

$$\tilde{x}(t, k) = x(t)$$

Proof:

- Construct a hybrid solution based on the dynamics of the timer τ
- Assumptions guarantee that the solution is complete

Hybrid ISS Lyapunov Function

An auxiliary **timer** $\tau \in [0, \Gamma]$ that models the constraints on the switching

$$\begin{cases} \dot{\tau} \in [0, \gamma], & \text{if } \tilde{\sigma} \in \mathcal{P}_s \\ \dot{\tau} = \gamma - (\lambda_s + \lambda_u), & \text{if } \tilde{\sigma} \in \mathcal{P}_u \end{cases}$$
$$\tau^+ = \tau - \ln(\mu)$$

A hybrid system with the state $z = (\tilde{x}, \tilde{\sigma}, \tau)$

A **hybrid ISS Lyapunov function** $V(z) := V_{\tilde{\sigma}}(\tilde{x})e^{\tau}$

Proposition 2

V decreases along all hybrid solutions to the hybrid system

Proposition 3

Every complete solution to the hybrid system satisfies

$$|\tilde{x}(t, k)| \leq \beta(|\tilde{x}(0, 0)|, t) + \alpha(\|\tilde{u}\|_{(t, k)})$$



Outline

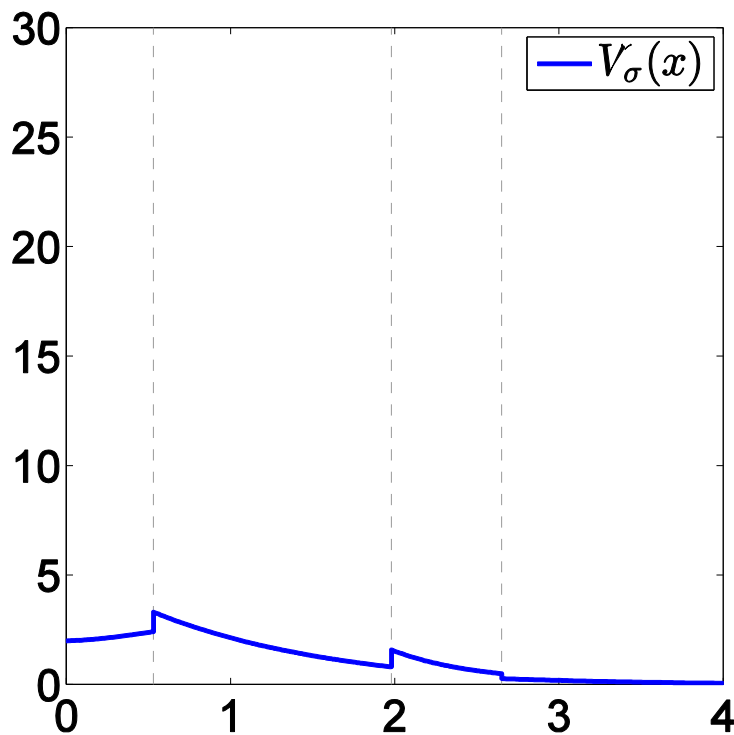
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Simulation

Subsystem 1

$$\dot{x}_1 = x_2,$$

$$\dot{x}_2 = -x_2 - x_1 + d.$$

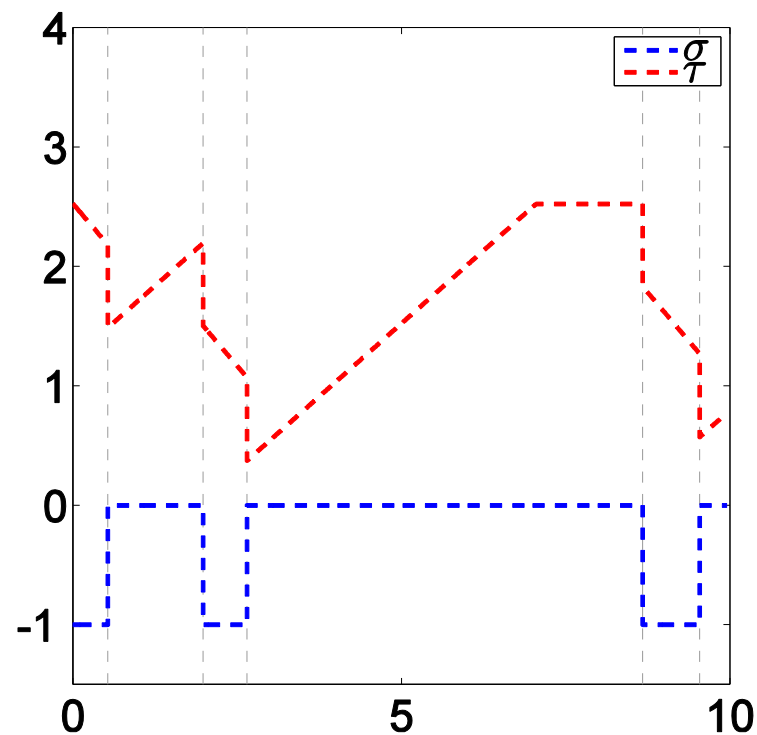


ISS Lyapunov functions

Subsystem 2:

$$\dot{x}_1 = x_2,$$

$$\dot{x}_2 = -x_2 + d.$$



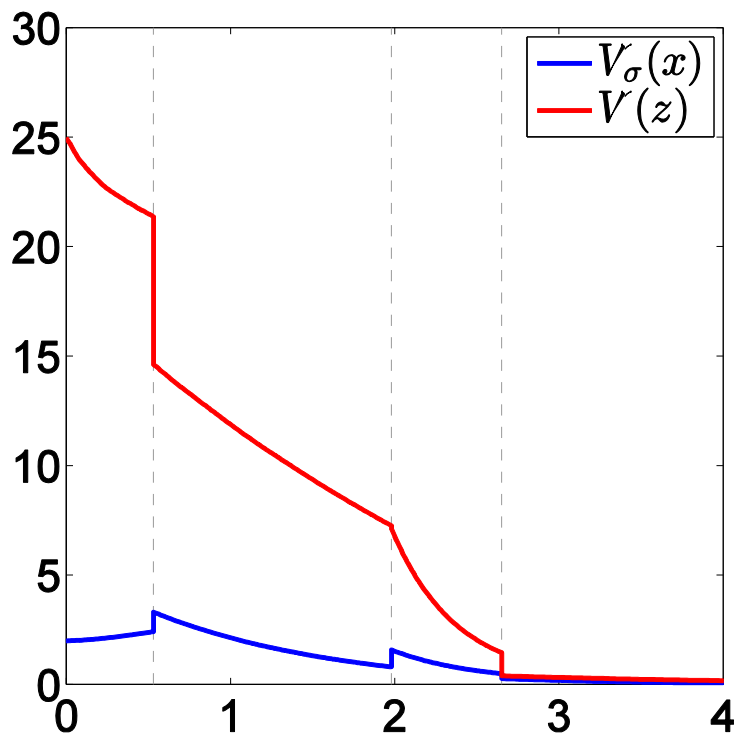
Switching signal and timer

Simulation

□ Subsystem 1

$$\dot{x}_1 = x_2,$$

$$\dot{x}_2 = -x_2 - x_1 + d.$$

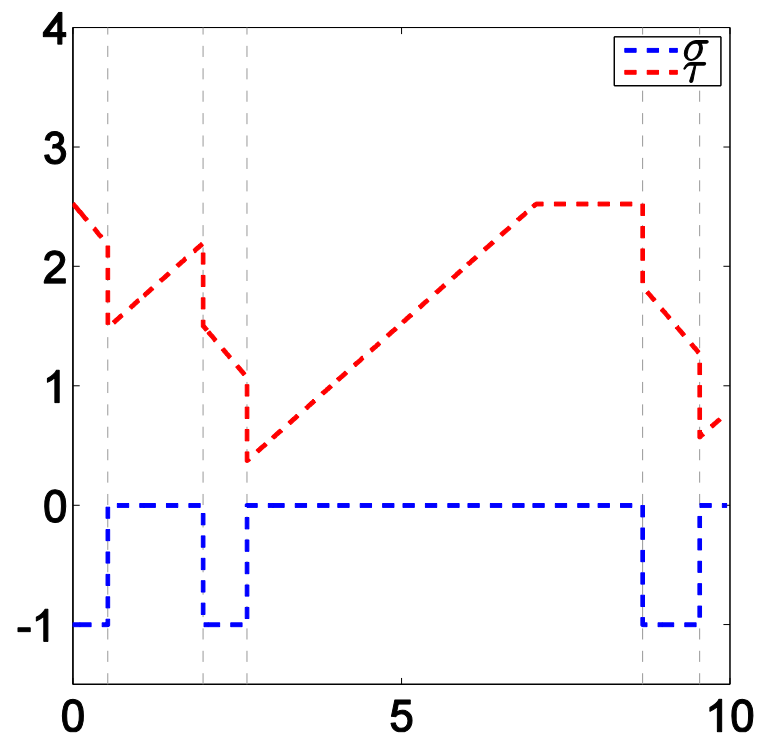


ISS Lyapunov functions

□ Subsystem 2:

$$\dot{x}_1 = x_2,$$

$$\dot{x}_2 = -x_2 + d.$$



Switching signal and timer

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Conclusion

Contribution

- Timer τ
 - Similar techniques in previous results:
 - [L-03] for switched system
 - [Hespanha-L-Teel-08] for impulsive system
 - [L-Nešić-Teel-14] for hybrid system
 - **Advantage:** our construction is able to handle both switches and non-ISS subsystems

Conclusion

Contribution

- Timer τ
- Hybrid ISS Lyapunov function $V(z) := V_{\tilde{\sigma}}(\tilde{x})e^{\tau}$
 - Application: **interconnected switched systems** [Y-L-SCL]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} f_{1,\sigma_1}(x_1, x_2) \\ f_{2,\sigma_2}(x_2, x_1) \end{bmatrix}$$

- For each switched system, construct a hybrid ISS Lyapunov function V_i , $i \in \{1, 2\}$
- Construction a Lyapunov function V of the interconnected system based on V_1, V_2
- Global asymptotic stability of the interconnected system

Conclusion

Summary

- ISS of a switched system with non-ISS subsystems
- Constraints on dynamics
- Constraints on switching
- Sufficient condition for ISS
- Auxiliary timer
- Hybrid ISS Lyapunov function

Thank you for your attention!
